

SQL Assignment Project

Retail Store Analytics System

(Database to GROUP BY & HAVING)

Objective

You are working as a Data Analyst in a retail company. The management wants to analyze customer purchases, product performance, sales trends, payment methods, and city-wise business performance using SQL.

Complete the following tasks step by step.

Step 1: Create Database

Create a database named:

retail_store_db

Use the database before creating any tables.

Step 2: Create Table

Create a table named `sales` with the following columns:

Column Name	Data Type
sale_id	INT
customer_name	VARCHAR(50)
gender	VARCHAR(10)
age	INT
city	VARCHAR(30)
category	VARCHAR(30)
product_name	VARCHAR(50)
payment_mode	VARCHAR(20)
quantity	INT
price	DECIMAL(10,2)
rating	DECIMAL(2,1)
order_date	DATE

Step 3: Insert Data

Insert the following records into the table.

INSERT INTO sales VALUES

(1,'Akash','Male',24,'Pune','Electronics','Laptop','UPI',1,55000,4.8,'2026-01-10'),
(2,'Rahul','Male',26,'Mumbai','Fashion','Shirt','Card',2,2500,4.2,'2026-01-11'),
(3,'Sneha','Female',23,'Delhi','Beauty','Lipstick','UPI',3,1500,4.5,'2026-01-12'),
(4,'Priya','Female',25,'Pune','Electronics','Mobile','Card',1,25000,4.7,'2026-01-13'),
(5,'Aman','Male',28,'Indore','Sports','Cricket Bat','Cash',1,3000,4.1,'2026-01-14'),
(6,'Rohit','Male',27,'Mumbai','Electronics','Headphones','UPI',2,4000,4.3,'2026-01-15'),
(7,'Riya','Female',24,'Delhi','Fashion','Jeans','Card',1,2200,4.4,'2026-01-16'),
(8,'Karan','Male',29,'Pune','Sports','Football','Cash',2,1800,4.0,'2026-01-17'),
(9,'Neha','Female',22,'Bhopal','Beauty','Face Wash','UPI',4,800,4.6,'2026-01-18'),
(10,'Vikas','Male',31,'Jaipur','Electronics','Tablet','Card',1,18000,4.5,'2026-01-19'),
(11,'Pooja','Female',27,'Pune','Fashion','Kurti','UPI',2,1700,4.3,'2026-01-20'),
(12,'Arjun','Male',30,'Delhi','Electronics','Smart Watch','Card',1,9000,4.6,'2026-01-21'),
(13,'Anjali','Female',26,'Mumbai','Beauty','Perfume','Cash',1,3500,4.2,'2026-01-22'),
(14,'Yash','Male',25,'Indore','Sports','Badminton Kit','UPI',1,2800,4.4,'2026-01-23'),
(15,'Meera','Female',28,'Pune','Electronics','Camera','Card',1,45000,4.9,'2026-01-24'),
(16,'Sahil','Male',24,'Bhopal','Fashion','Jacket','UPI',1,3200,4.1,'2026-01-25'),
(17,'Nikita','Female',29,'Jaipur','Beauty','Moisturizer','Card',2,1200,4.7,'2026-01-26'),
(18,'Harsh','Male',32,'Delhi','Sports','Gym Gloves','Cash',3,900,4.0,'2026-01-27'),
(19,'Simran','Female',23,'Mumbai','Electronics','Earbuds','UPI',2,2500,4.5,'2026-01-28'),
(20,'Deepak','Male',27,'Pune','Fashion','T-Shirt','Card',3,1800,4.2,'2026-01-29');

Part A – Basic SQL Queries

Q1. Display all records from the sales table.

Q2. Display only customer names.

Q3. Display customer names and cities.

Q4. Display product names and prices.

Q5. Display all unique cities.

Q6. Display all unique categories.

Q7. Display all unique payment methods.

Q8. Display all male customers.

Q9. Display all female customers.

Q10. Display customer name and rating.

Part B – WHERE Clause

Q11. Display all customers from Pune.

Q12. Display all customers from Mumbai.

Q13. Display all Electronics products.

Q14. Display all Fashion products.

Q15. Display all Sports products.

Q16. Display products with price greater than 5000.

Q17. Display products with price less than 5000.

Q18. Display customers whose age is greater than 25.

Q19. Display customers whose age is less than 25.

Q20. Display products with rating greater than 4.5.

Q21. Display all UPI transactions.

Q22. Display all Card transactions.

Q23. Display all Cash transactions.

Q24. Display Electronics products from Pune.

Q25. Display Fashion products from Mumbai.

Q26. Display Female customers from Delhi.

Q27. Display Male customers from Pune.

Q28. Display products priced between 2000 and 10000.

Q29. Display customers aged between 24 and 30.

Q30. Display orders placed after 20 January 2026.

Part C – ORDER BY, LIMIT and OFFSET

Q31. Sort products by price in ascending order.

Q32. Sort products by price in descending order.

Q33. Sort customers by age in ascending order.

Q34. Sort customers by age in descending order.

Q35. Sort products by rating in descending order.

- Q36. Display the highest priced product.**
- Q37. Display the lowest priced product.**
- Q38. Display the highest rated product.**
- Q39. Display the latest order.**
- Q40. Display the oldest order.**
- Q41. Display the top 5 records.**
- Q42. Display the top 3 most expensive products.**
- Q43. Display the top 5 highest rated products.**
- Q44. Display the second highest priced product.**
- Q45. Display the third highest priced product.**
- Q46. Skip the first 5 rows and display the next records.**
- Q47. Display rows 6 to 10.**
- Q48. Display rows 11 to 15.**
- Q49. Display rows 16 to 20.**
- Q50. Display the top 3 latest orders.**
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Part D – Aggregate Functions

- Q51. Find the total number of orders.**
- Q52. Find the total sales revenue.**
- Q53. Find the average sales amount.**

- Q54. Find the maximum sales amount.**
- Q55. Find the minimum sales amount.**
- Q56. Find the highest rating.**
- Q57. Find the lowest rating.**
- Q58. Find the average rating.**
- Q59. Find the total quantity sold.**
- Q60. Find the average quantity sold.**
- Q61. Find the maximum quantity sold.**
- Q62. Find the minimum quantity sold.**
- Q63. Count the number of Electronics products.**
- Q64. Count the number of Fashion products.**
- Q65. Count the number of Sports products.**
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Part E – String Functions

- Q66. Convert all customer names to uppercase.**
- Q67. Convert all customer names to lowercase.**
- Q68. Find the length of each customer name.**
- Q69. Find the length of each city name.**
- Q70. Combine customer name and city in one column.**
- Q71. Replace the letter 'a' with '@' in customer names.**

Q72. Replace the letter 'e' with '*' in customer names.

Q73. Remove extra spaces using TRIM().

Q74. Display the first 3 letters of each customer name.

Q75. Display the last 2 letters of each city name.

Part F – Numeric and Date Functions

Q76. Round all prices.

Q77. Apply CEIL() on prices.

Q78. Apply FLOOR() on prices.

Q79. Find MOD(quantity,2).

Q80. Find MOD(quantity,3).

Q81. Display the current date.

Q82. Display the current time.

Q83. Display the current timestamp.

Q84. Extract the year from order_date.

Q85. Extract the month from order_date.

Q86. Extract the day from order_date.

Q87. Display orders placed in January.

Q88. Display orders placed before 20 January 2026.

Q89. Display orders placed after 20 January 2026.

Q90. Display the latest order date.

Part G – UPDATE and DELETE

Q91. Update the city of Akash.

Q92. Update the rating of Rahul.

Q93. Change all Card payments to UPI.

Q94. Increase all product prices by 500.

Q95. Decrease all product prices by 100.

Q96. Change category Electronics to Gadgets.

Q97. Change city Pune to Nagpur.

Q98. Update the quantity of a selected product.

Q99. Update a product name.

Q100. Update a customer name.

Q101. Delete a specific customer record.

Q102. Delete all Sports category records.

Q103. Delete records with rating below 4.0.

Q104. Delete all Cash payment records.

Q105. Delete records from Jaipur.

Part H – GROUP BY

- Q106. Display city-wise order count.**
 - Q107. Display category-wise order count.**
 - Q108. Display payment mode-wise order count.**
 - Q109. Display gender-wise order count.**
 - Q110. Display city-wise total revenue.**
 - Q111. Display category-wise total revenue.**
 - Q112. Display payment mode-wise total revenue.**
 - Q113. Display city-wise average revenue.**
 - Q114. Display category-wise average revenue.**
 - Q115. Display city-wise maximum sale.**
 - Q116. Display city-wise minimum sale.**
 - Q117. Display category-wise maximum sale.**
 - Q118. Display category-wise minimum sale.**
 - Q119. Display city and category-wise revenue.**
 - Q120. Display city and payment mode-wise revenue.**
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Part I – HAVING Clause

- Q121. Display cities having total revenue greater than 50,000.**
- Q122. Display categories having total revenue greater than 60,000.**
- Q123. Display cities having average sale greater than 10,000.**

- Q124. Display categories having average rating greater than 4.5.**
- Q125. Display cities having more than 3 orders.**
- Q126. Display categories having more than 2 orders.**
- Q127. Display payment modes having more than 5 orders.**
- Q128. Display cities having maximum sale greater than 40,000.**
- Q129. Display categories having minimum sale greater than 1,500.**
- Q130. Display cities having average age greater than 25.**
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Part J – WHERE + GROUP BY + HAVING

- Q131. Display city-wise revenue for UPI transactions.**
- Q132. Display city-wise revenue for Electronics products.**
- Q133. Display city-wise revenue generated by female customers.**
- Q134. Display city-wise revenue for orders above 5,000.**
- Q135. Display cities where revenue from orders above 5,000 exceeds 50,000.**
- Q136. Display categories with more than 2 UPI orders.**
- Q137. Display categories where Card payment revenue exceeds 10,000.**
- Q138. Display category-wise count of products having rating above 4.5.**
- Q139. Display city-wise revenue generated by customers older than 25.**
- Q140. Display cities where revenue generated by customers older than 25 exceeds 20,000.**
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Final Business Reports

Q141. Identify the city generating the highest revenue.

Q142. Identify the category generating the highest revenue.

Q143. Find the most frequently used payment method.

Q144. Find the best-rated category.

Q145. Find the highest spending customer.

Q146. Find the most popular product.

Q147. Display the top 3 cities by revenue.

Q148. Display the top 3 categories by revenue.

Q149. Generate a monthly sales summary report.

Q150. Generate a complete business performance report using all concepts learned so far.